

Product Requirements Document: InsureFlow AI

Version: 1.0 **Status:** Draft **Technology Stack:** Built via Claude Code (AI-Powered)

1. Executive Summary

InsureFlow AI is a high-speed validation layer designed for hospital insurance desk agents. The app's core mission is to bridge the gap between hospital documentation and TPA (Third-Party Administrator) requirements. By using a **"Repair, don't Reject"** philosophy, the app ensures that every insurance claim or cashless request is 100% compliant before it ever reaches the insurer.

2. Product Objectives

- **Zero Technical Rejections:** Eliminate rejections caused by missing signatures, blurry scans, or incomplete forms.
 - **Clinical Alignment:** Use AI to highlight medical necessity, ensuring the doctor's notes justify the billed procedure.
 - **Operational Speed:** Reduce the "back-and-forth" between hospitals and TPAs from days to minutes.
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3. User Experience (UX) Flow

The app follows a "Single-Stream" intake process:

1. **Intake:** Agent uploads a patient ID/Insurance card via mobile camera or gallery.
 2. **ID Generation:** System instantly generates a **Unique Claim ID**.
 3. **Bulk Upload:** Agent uploads all documents (bills, discharge summaries, reports) in one go.
 4. **Live Validation:** AI scans for signatures and data consistency.
 5. **Final Output:** A bookmarked, indexed "Master PDF" is generated for TPA submission.
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4. Functional Requirements

Phase 1: POC (Proof of Concept)

Feature ID	Feature Name	Description
FR-1.1	Unique ID Engine	Instant generation of Claim IDs for session tracking.
FR-1.2	Signature Detector	OCR logic to detect presence/absence of doctor stamps and signs.
FR-1.3	Basic Intake UI	Simple "Drop Zone" for files and camera integration.

Phase 2: First Release (The Validation Gate)

Feature ID	Feature Name	Description
FR-2.1	Plan Logic Engine	Scrapes public insurance data to check plan-specific requirements.
FR-2.2	Clinical Highlighting	Highlights medical necessity keywords in RED to assist TPA review.
FR-2.3	Repair & Reject UX	A dashboard that suggests specific "repairs" rather than rejecting the file.
FR-2.4	Master PDF Tool	Merges and auto-indexes documents into a professional submission packet.

Phase 3: Future Enhancements (Next Release)

Feature ID	Feature Name	Description
FR-3.1	Verification Cover Page	A digital "Seal of Approval" certifying the pre-audit results.
FR-3.2	Anti-Blur Guardrail	Real-time camera feedback to prevent poor quality uploads.
FR-3.3	TPA Risk Score	Predictive AI score indicating the likelihood of immediate approval.

5. Non-Functional Requirements

- **Security:** HIPAA-compliant data handling and encryption at rest.
- **Performance:** Document validation (OCR/Analysis) must complete in < 15 seconds.
- **Reliability:** 99.9% uptime for the public insurance plan database.

6. Website Structure (Professional Presence)

1. **Home Page:** Focus on "Zero-Rejection Filing."
2. **The Tech:** Explaining the "Three Category" validation (Demographics, Medical, Billing).
3. **TPA Portal:** Informational page for insurers on why InsureFlow data is superior.
4. **Security/Compliance:** Detailed overview of data protection.
5. **Contact/Demo:** Lead generation for hospital administrators.